
K-Means++ Numerical Example

We want **k = 2 clusters**.

We have **6 data points** on a 1-D line (to keep it very simple):

Point	Value
X ₁	2
X ₂	4
X ₃	5
X ₄	8
X ₅	12
X ₆	14

STEP 1 — Choose first centroid randomly

Suppose randomly selected:

First centroid (C_1) = 5

STEP 2 — Compute distance² of each point to nearest chosen centroid

Distance from each point to $C_1 = 5$:

Point	Value	Distance	Distance ²
X ₁	2		$5 - 2$

Point	Value	Distance	Distance ²
X ₂	4		5 – 4
X ₃	5	0	0
X ₄	8		8 – 5
X ₅	12		12 – 5
X ₆	14		14 – 5

STEP 3 — Compute probabilities for selecting next centroid

Formula:

$$P(i) = \frac{D_i^2}{\sum D_i^2}$$

Total:

$$\sum D^2 = 9 + 1 + 0 + 9 + 49 + 81 = 149$$

Now probabilities:

Point	D ²	Probability P(i)
X ₁	9	9/149 = 0.060
X ₂	1	1/149 = 0.006
X ₃	0	0
X ₄	9	9/149 = 0.060
X ₅	49	49/149 = 0.329
X ₆	81	81/149 = 0.544

STEP 4 — Choose second centroid based on probabilities

Highest probabilities are:

- $x_6(14) \rightarrow 0.544$
- $x_5(12) \rightarrow 0.329$

So chances are high second centroid will be chosen near the far end.

Assume we randomly pick based on distribution:

Second centroid (C_2) = 14

Final K-Means++ chosen initial centroids:

$$C_1=5, C_2=14$$

These are **well-spread**, which is the purpose of K-Means++.

Data (1-D, $k=2$)

Points: 2, 4, 5, 8, 12, 14

Initial centroids (from K-Means++ step)

$$C_1=5, C_2=14$$

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Iteration 1 — Assignment step

Compute absolute distance of each point to each centroid and assign to nearest.

- Point 2: $|2 - 5| = 3$ vs $|2 - 14| = 12 \rightarrow$ **cluster 1**
- Point 4: $|4 - 5| = 1$ vs $|4 - 14| = 10 \rightarrow$ **cluster 1**
- Point 5: $|5 - 5| = 0$ vs $|5 - 14| = 9 \rightarrow$ **cluster 1**
- Point 8: $|8 - 5| = 3$ vs $|8 - 14| = 6 \rightarrow$ **cluster 1**
- Point 12: $|12 - 5| = 7$ vs $|12 - 14| = 2 \rightarrow$ **cluster 2**
- Point 14: $|14 - 5| = 9$ vs $|14 - 14| = 0 \rightarrow$ **cluster 2**

So:

- Cluster 1 = $\{2, 4, 5, 8\}$
- Cluster 2 = $\{12, 14\}$

Update centroids (means)

- New $C_1 = \frac{2+4+5+8}{4} = \frac{19}{4} = 4.75$
 - New $C_2 = \frac{12+14}{2} = 13$
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Iteration 2 — Assignment step (with $C_1 = 4.75, C_2 = 13$)

Distances:

- Point 2: $|2 - 4.75| = 2.75$ vs $|2 - 13| = 11 \rightarrow$ **cluster 1**
- Point 4: $|4 - 4.75| = 0.75$ vs $|4 - 13| = 9 \rightarrow$ **cluster 1**
- Point 5: $|5 - 4.75| = 0.25$ vs $|5 - 13| = 8 \rightarrow$ **cluster 1**
- Point 8: $|8 - 4.75| = 3.25$ vs $|8 - 13| = 5 \rightarrow$ **cluster 1**

- Point 12: $|12 - 4.75| = 7.25$ vs $|12 - 13| = 1 \rightarrow$ **cluster 2**
- Point 14: $|14 - 4.75| = 9.25$ vs $|14 - 13| = 1 \rightarrow$ **cluster 2**

Assignments remain:

- Cluster 1 = $\{2, 4, 5, 8\}$
- Cluster 2 = $\{12, 14\}$

Update centroids again:

- $C_1 = 4.75$ (same)
 - $C_2 = 13$ (same)
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Convergence

Since cluster assignments (and centroids) did **not change** from Iteration 1 $\rightarrow$ Iteration 2, the algorithm has converged.

Final result

- **Cluster A:** points $\{2, 4, 5, 8\}$ with centroid 4.75
- **Cluster B:** points $\{12, 14\}$ with centroid 13

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Results

- SSE (cluster 1, centroid = 4.75) = **18.75**
- SSE (cluster 2, centroid = 13.0) = **2.00**
- **Total SSE = 20.75**